

Welfare, Integrability and Afriat

Advanced Microeconomics 1: Economic Decision Making. Lecture 4

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Where we left the question

The program so far, and the one gap left in it:

1. **L1**: the question stated; rationality testable; basket indexation over-compensates.
2. **L2**: the ideal answer defined — $e(p^1, u^0) - w$, per household.
3. **L3**: the answer made estimable — areas under Hicksian curves; bias $= \Delta p^T S \Delta p / 2$.
4. **Today**: the missing license. Every bridge assumed a rational household *behind* the data. When does observed demand actually **pin down** preferences? Then: execute the answer on Belgium's index — and present **the bill** for every assumption used.

Invoice items 4 and 6: recovery, then audit.

Recovery I: demand functions (integrability)

The integrability problem

We know the forward direction: rational preferences $\Rightarrow x(p, w)$ satisfies Walras' law, h.d.0, and S symmetric + NSD.

The reverse question

Given a demand function $x(p, w)$ with those properties — does a rational preference relation exist that *generates* it?

Theorem · integrability (MWG 3.H)

Yes. If $x(p, w)$ satisfies Walras' law, h.d.0, and has a symmetric, negative semidefinite Slutsky matrix, then there exists a rational preference relation rationalizing x . **Proof**

Proof: constructive sketch in the appendix; full treatment MWG 3.H — read it once, if only to realize the beauty of the theory.

Four consequences, in rising order of importance for us:

1. Walras, h.d.0, symmetry, NSD are the *only* consequences of preference maximization for demand functions. The theory's complete fingerprint — four checkable conditions.
2. Symmetry is the *only* content beyond WARP's implications. Lecture 1's gap, measured exactly: one matrix condition.
3. An empirical strategy falls out: posit a tractable demand system, estimate it, *test* the restrictions (this is what Deaton–Muellbauer-style work does).
4. **Welfare analysis is licensed**: preferences can be *recovered* from observed demand — so e , CV, EV are computable from data. Without this theorem, Lectures 2–3 were a castle in the air. With it, the compensation program stands.

Recovery II: finite data (SARP and Afriat)

Finite data: strengthening WARP to SARP

An estimated demand *function* is a luxury. Often we hold finitely many observations — or 25 quiz choices.

Definition · SARP (Houthakker 1950)

$x(p, w)$ satisfies **SARP** if for any chain $(p^1, w^1), \dots, (p^N, w^N)$ with $x(p^{n+1}, w^{n+1}) \neq x(p^n, w^n)$ and $p^n \cdot x(p^{n+1}, w^{n+1}) \leq w^n$ for all $n \leq N - 1$, we have $p^N \cdot x(p^1, w^1) > w^N$.

Theorem · Houthakker (1950)

$x(p, w)$ satisfies SARP $\iff$ there is a rational preference relation that rationalizes x .

Proof deferred (Houthakker 1950; MWG 3.J). Chains may never cycle back — L1's three-good disease, outlawed at every length; equivalent to Slutsky symmetry + negativity. For $L = 2$: WARP $\iff$ SARP (MWG Ex. 3.J.1, a good self-test) — why the two-good quiz needed only the weak axiom.

Afriat's theorem: Monday's practical, in advance

Finite data $\{(p^t, x^t)\}_{t=1}^T$; GARP as in Lecture 1:

Theorem · Afriat (1967)

The following are equivalent:

- (i) the data satisfy GARP;
- (ii) there exist $U_t, \lambda_t > 0$ with $U_s \leq U_t + \lambda_t p^t \cdot (x^s - x^t) \forall s, t$ (*Afriat inequalities*);
- (iii) the data are rationalizable by a continuous, increasing, **concave** utility function.

Proof

- For finite data, utility maximization *is* GARP. Concavity comes free: finite data can never reject it.
- **Monday, previewed**: you will code (i) as a GARP check on this room's quiz choices; the CCEI — the largest budget deflator at which (i) holds — scores how close each of you comes. Benchmark: CKMS adult median CCEI ≈ 0.9 [verify exact figure].

The answer, executed

Question (a) and (b), answered rung by rung — with the assumption each rung stands on:

1. **Arithmetic** [no assumptions]: quartile weights $\times$ published prices $= \pi^h$. Gas + electricity alone: 7.2pp of Q1's 2022 inflation vs 3.9pp of Q4's (NBB): a 3.3pp gap from two goods.
2. **Revealed preference** [choice reveals a stable “better off”]: basket compensation $\geq$ the true cost of staying as well off (L1, three lines). Sign of the error: over-pays — NBB Box 2, in print.
3. **Rationality + continuity** [a ranking exists and u represents it]: the exact bill $e^h(p^1, u^0) - w$ is *defined*; integrability/Afriat [symmetry / GARP] license recovering it from data.
4. **Calibration** [+ Cobb–Douglas shares]: the bill computed — next slide.
5. **Full baskets** [+ estimated demand; non-homothetic correction]: CLO's UK accounting — substitution 1.5pp of 26.2%; distribution 7.4pp.

Rung 4: the Belgian bill, computed

goal (b): policy

Home energy +57.9% over 2022 (NBB); home-energy budget shares 10% (poorest quartile) vs 6% (richest). Cobb–Douglas within (energy, everything else):

$$\frac{-CV}{w} = \left(\frac{p_E^1}{p_E^0} \right)^\alpha - 1 = 1.579^\alpha - 1, \quad \frac{\Delta w_{\text{basket}}}{w} = \alpha \times 57.9\%.$$

	exact bill $-CV/w$	basket pays	over-pay
poorest quartile, $\alpha = 0.10$	4.7%	5.8%	1.1 pp
richest quartile, $\alpha = 0.06$	2.8%	3.5%	0.7 pp

- Each cell's warrant: the formula is R8–R9 (duality, e); the column ranking is R2/R12 (Konüs $\leq$ Laspeyres); the *numbers* add the Cobb–Douglas approximation (constant shares; EC1 P8 shows what breaks when they move).
- So: the right payments differ by 1.9pp of budget across quartiles, and the single-basket scheme over-pays both — more, in absolute terms, where it matters least.

Hausman (AER 1981), the block's (b)-methods paper, run on the thread:

1. **Estimate** one Marshallian demand: $x_E(p_E, w)$ for energy, from budget-survey data.
2. **Integrate**: Roy's identity, read backwards, is an ordinary differential equation in v ; solve it along the price path. One estimated curve pins down indirect utility *exactly* — no functional leap of faith beyond the demand specification itself.
3. **Invert**: $e = v^{-1}$ in the wealth argument. Read off $e(p^1, u^0) - w$: the exact CV — and the deadweight loss of any tax on the good, exactly, not by triangle approximation.

Every step is a Lecture 3 bridge earning its keep: Roy backwards (step 2), duality (step 3), and integrability standing behind the whole thing — the license to treat the estimated x as *someone's* demand.

Hold the machinery up against the actual mechanism:

- **Fixed basket:** the health index is Laspeyres-type $\Rightarrow$ over-states the true rise by $\approx |\Delta p^T S \Delta p|/2$. Measured through a surge: UK groceries 2021Q3–2023Q3, Laspeyres 26.2% vs. true cost-of-living 24.7% — 1.5pp of substitution in nine quarters (Chen–Levell–O’Connell 2025). [Boskin’s US commission: total CPI bias ~ 1.1 pp/yr — verify before quoting.]
- **Exclusions:** the health index drops motor fuels, alcohol, tobacco. Whatever the rationale, excluding goods from the compensation index is *a welfare judgment* — it declares those price rises not owed to you. The theory makes the judgment visible; it cannot make it for you.
- **The central bank agrees:** NBB Annual Report 2022, Box 2 — published 2022 inflation “led to *overcompensation for the average household*.” Lecture 1’s three-line theorem, in an official document, with a second mechanism attached (new-contract energy pricing) that pushes the same way.

One number for everyone: who wins, who loses

goal (b): policy

The index is plutocratic (expenditure-weighted: each euro votes, so rich baskets count more) and unique (one π for all h). Against the machinery:

household	true cost change vs. index	indexation outcome
low income, no social tariff	$\pi^h > \pi$ (energy-heavy budget)	under -compensated
low income, social tariff	shielded on energy prices	closer to whole
average household	$\pi^h \approx \pi$, minus substitution	over -compensated (NBB)
high income	$\pi^h < \pi$; fast-indexing sectors	over -compensated
annual-January workers	correct π , one year late	real loss in the gap year
self-employed	no automatic indexation	bear the shock

Question (b), answered as far as theory plus official accounting can take it: **indexation protects the average and misses the tails — in both directions**. The tails are wide: in the UK surge, bottom-decile inflation ran 7.4pp above top-decile, and substitution offset less of the rise for the poorest quartile (5.5%) than the richest (6.9%) (Chen–Levell–O’Connell 2025).

The bill

What the euro answer leaned on

Every number this block produced rests on four load-bearing assumptions. Time to audit each:

1. **Rationality** — choices reveal a stable ranking. *Tested:* GARP, CCEI. On you.
2. **No money illusion** — h.d.0, filed away on the first content slide of Lecture 1.
3. **Stable preferences** — the u^0 being restored is the same object across the comparison.
4. **A choice of reference** — CV or EV; we treated it as bookkeeping.

None of these is safe. The next three slides present the evidence — starting with data you generated.

Exhibit A: your own answers

Week 1, two quiz questions, same object:

- “You own the mug — minimum price to sell?” [insert: class median WTA]
- “You don’t — maximum price to buy?” [insert: class median WTP]
- Under assumptions 1–4, for a small-budget-share good, these should nearly coincide. [insert: class ratio.] The canonical finding: $WTA \approx 2 \times WTP$.

The block’s third paper, revealed

Kahneman, Knetsch & Thaler (JPE 1990), “Experimental Tests of the Endowment Effect and the Coase Theorem” — mugs, real money, the gap survives markets and experience. Owning the reference point moves the valuation.

Consequence for us: CV and EV are elicited from different reference points — so they can differ *far* beyond wealth effects. “Which number does policy owe?” stops being bookkeeping.

Exhibit B: money illusion

Assumption 2 — h.d.0, “only relative prices and real wealth matter” — has its own file:

- Shafir, Diamond & Tversky (QJE 1997): in surveys and experiments, people systematically evaluate transactions partly in *nominal* terms — a 2% raise under 4% inflation feels better than a 2% cut under zero, same real outcome.
- Precisely the environment where it matters is ours: **inflations**. Wage bargaining, rent adjustment, the politics of indexation itself run partly on nominal frames — one reason indexation is beloved where it exists and resisted where it does not.
- Note the discipline: L1 *filed* the assumption instead of burying it; the theory told us exactly where to look for the crack. That is what “testable assumptions” buys.

The bill, itemized: if this fails, the answer errs that way

assumption at risk	evidence	direction of error
substitution ignored (basket)	theorem (L1) + NBB	over-pays; worst in relative shocks
one index, many π^h	Jaravel; NBB quartiles	under-pays poor, over-pays rich
rationality (GARP, CCEI)	CKMS; Monday, this room	recovery unlicensed where it fails
no money illusion	SDT 1997	“real” compensation mis-perceived
homotheticity (trading down)	CLO 2025; EC1 P8	superlative tracks neither Konüs bound
reference independence	KKT 1990	elicited answer depends on the frame

- The point is not nihilism. Each row is a *signed, bounded* correction — the theory that produced the answer also produces its error analysis. That is the difference between a model and an opinion.

The block's scorecard

Takeaways

- **(a) Who did the 2022 shock hurt, and by how much?** Answered: π^h from shares alone; exact per-household CV via recovery + Hausman's pipeline; validity certified household-by-household by GARP/CCEI — including yours, on Monday.
- **(b) Is one-basket indexation the right compensation?** No — it over-pays the average (theorem; NBB in print) and misses both tails (quartile weights; Jaravel). The right payment is $e^h(p^1, u^0) - w$, per household — and we can compute it.
- **The bill:** reference dependence, money illusion and failed rationality each bend the answer in a known direction. Use the numbers; carry the error bars.

Revealed reading: KKT (JPE 1990); Shafir, Diamond & Tversky (QJE 1997); Jaravel (QJE 2019) in full.

Ahead: the practical on Monday (laptop, working R, your quiz ID); EC1 on Thursday; then the firms block — where the prices that hurt households come *from*.

Appendix: proofs

Proof sketch: integrability (MWG 3.H)

Step 1 — recover e from x . Fix (p^0, w^0) and try to solve the system of PDEs given by Shephard run backwards:

$$\nabla_p e(p) = x(p, e(p)), \quad e(p^0) = w^0.$$

A solution exists locally iff the system is consistent — Frobenius' condition: the “cross-derivatives” of the right-hand side must commute. Computing them, the condition is exactly **symmetry of $S(p, e(p))$** . That is the moment symmetry stops being decoration: it is the solvability condition for reconstructing the cost of living from demand.

Step 2 — recover preferences from e . Given $e(\cdot, u)$ (concave, by NSD, in the recovered system), define upper contour sets as the intersections of half-spaces

$$V_u = \{x \in \mathbb{R}_+^L : p \cdot x \geq e(p, u) \text{ for all } p \gg 0\}.$$

Concave-duality (support-function) arguments show the preference relation these sets induce is rational and generates x . $\square$ (sketch; full argument MWG 3.H.1–3.H.2)

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Proof sketch: Afriat's theorem

(iii) $\Rightarrow$ (i): a rationalizing utility orders the data; revealed-preference chains inherit the order, so no cycle can end strictly above its start.

(i) $\Rightarrow$ (ii): GARP makes the revealed relation acyclic, so observations can be ordered; a combinatorial induction then picks U_t , $\lambda_t > 0$ satisfying the inequalities — read as “a concave function with slope $\lambda_t p^t$ at x^t leaves room for U_s at x^s .”

(ii) $\Rightarrow$ (iii): build the utility explicitly as the lower envelope of those supporting hyperplanes:

$$u(x) = \min_t \{ U_t + \lambda_t p^t \cdot (x - x^t) \}$$

— continuous, increasing, concave by construction, and it rationalizes every observation.



Sketch; Afriat (1967), Varian (1982). The CCEI relaxes (i): your Thursday number is the largest budget scaling $e^* \leq 1$ at which GARP holds.